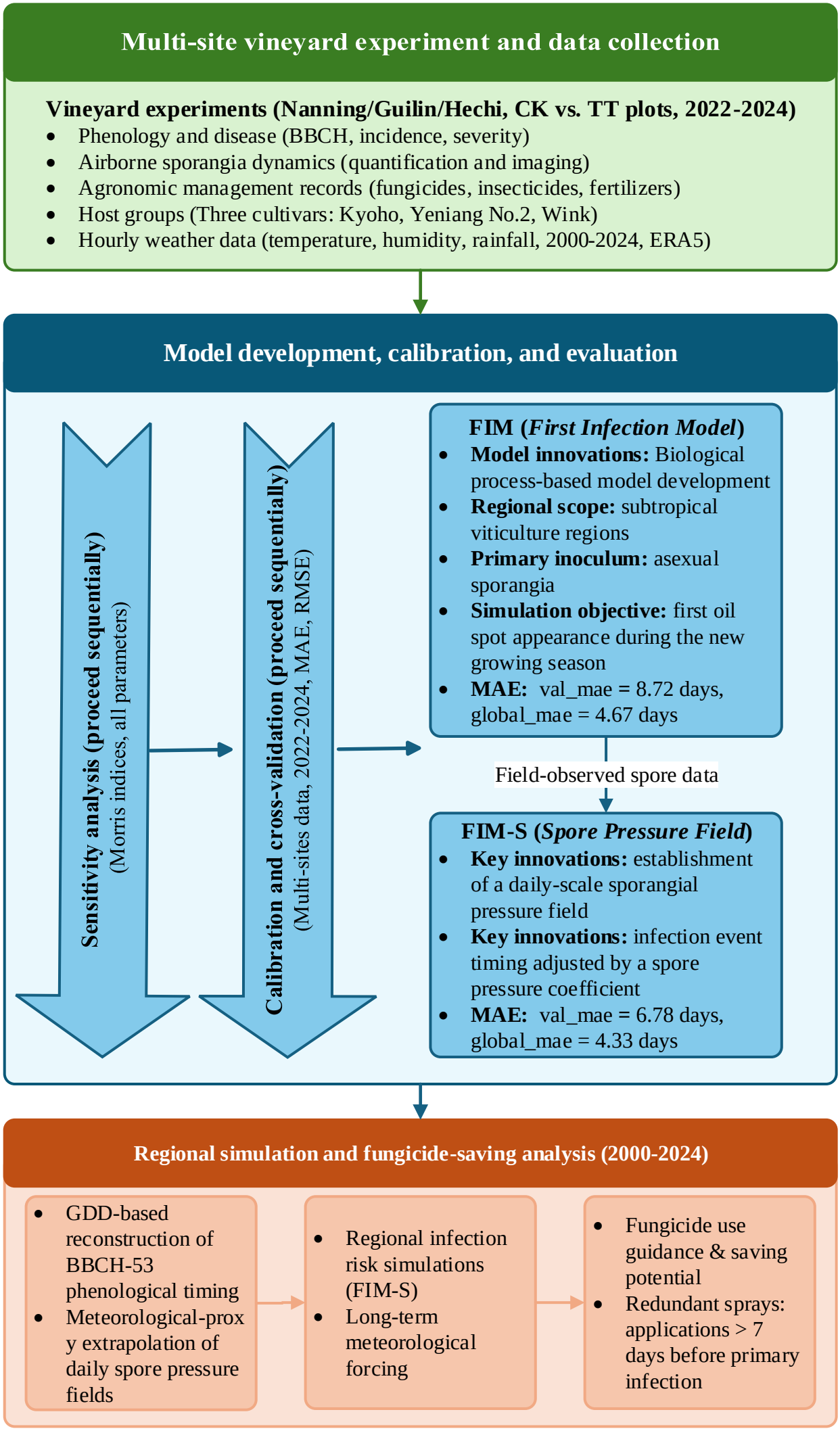

Multi-site vineyard experiment and data collection
Vineyard experiments (Nanning/Guilin/Hechi, CK vs. TT plots, 2022-2024)
Phenology and disease (BBCH, incidence, severity)
Airborne sporangia dynamics (quantification and imaging)
Agronomic management records (fungicides, insecticides, fertilizers)
Host groups (Three cultivars: Kyoho, Yeniang No.2, Wink)
Hourly weather data (temperature, humidity, rainfall, 2000-2024, ERA5)
Model development, calibration, and evaluation
Sensitivity analysis (proceed sequentially)
(Morris indices, all parameters)
Calibration and cross-validation (proceed sequentially)
(Multi-sites data, 2022-2024, MAE, RMSE)
FIM (First Infection Model)
Model innovations: Biological process-based model development
Regional scope: subtropical viticulture regions
Primary inoculum: asexual sporangia
Simulation objective: first oil spot appearance during the new growing season
MAE: val_mae = 8.72 days, global_mae = 4.67 days
Field-observed spore data
FIM-S (Spore Pressure Field)
Key innovations: establishment of a daily-scale sporangial pressure field
Key innovations: infection event timing adjusted by a spore pressure coefficient
MAE: val_mae = 6.78 days, global_mae = 4.33 days
Regional simulation and fungicide-saving analysis (2000-2024)
GDD-based reconstruction of BBCH-53 phenological timing
Meteorological-proxy extrapolation of daily spore pressure fields
Regional infection risk simulations (FIM-S)
Long-term meteorological forcing
Fungicide use guidance & saving potential
Redundant sprays: applications > 7 days before primary infection